



Chapter 1

The Foundations: Logic and Proofs

MAD101

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Proposition

1 Propositional Logic

Definition

A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either **true or false**, but not both.

Example.

1. Hà Nội is the capital of Việt Nam.
2. All prime numbers are even.
3. It is going to rain!
4. What times is it?



Proposition

1 Propositional Logic

Definition

A proposition is a declarative sentence (that is, a sentence that declares a fact) that is either **true or false**, but not both.

Example.

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2. All prime numbers are even.
3. It is going to rain!
4. What times is it?

Truth table

The truth value of a proposition is true, denoted by T, if it is a true proposition, and the truth value of a proposition is false, denoted by F, if it is a false proposition.

p
True/T/1
False/F/0

Logical operators - Negation

1 Propositional Logic

Definition

Let p be a proposition. The negation of p is the statement “It is not the case that p ”.

Notation: $\neg p$, $\bar{p}$, $\tilde{p}$, $-p$, p' , Np , and $!p$

p	$\neg p$
1	0
0	1

Example. Find the negation of the proposition “Michael’s PC runs Linux”.

Logical operators - Negation

1 Propositional Logic

Definition

Let p be a proposition. The negation of p is the statement “It is not the case that p ”.

Notation: $\neg p$, $\bar{p}$, $\tilde{p}$, $-p$, p' , Np , and $!p$

p	$\neg p$
1	0
0	1

Example. Find the negation of the proposition “Michael’s PC runs Linux”.

Solution. “Michael’s PC does not run Linux.”

Logical operators - Conjunction (hội)

1 Propositional Logic

Definition

Let p and q be propositions. The conjunction of p and q , denoted by $p \wedge q$, is the proposition “ p and q ”. The conjunction $p \wedge q$ is true when both p and q are true and is false otherwise.

p	q	$p \wedge q$
1	1	1
1	0	0
0	1	0
0	0	0
Chỉ đúng khi cả 2 cùng đúng		

Example 1. “He is young (p) and strong (q).” (In symbols: $p \wedge q$)

Example 2. “It is below freezing (p) but not snowing (q).” (In symbols: $p \wedge q$)

Logical operators - Disjunction (tuyển)

1 Propositional Logic

Definition

Let p and q be propositions. The disjunction of p and q , denoted by $p \vee q$, is the proposition “ p or q ”. The disjunction $p \vee q$ is false when both p and q are false and is true otherwise.

p	q	$p \vee q$
1	1	1
1	0	1
0	1	1
0	0	0
Chỉ sai khi cả 2 cùng sai		

Example. “ $3 > 2$ (p) or $2 > 5$ (q).” (In symbols: $p \vee q$)

Logical operators - Exclusive or (tuyển loại trừ)

1 Propositional Logic

Definition

Let p and q be propositions. The exclusive or of p and q , denoted by $p \oplus q$ (or p XOR q), is the proposition that is true when exactly one of p and q is true and is false otherwise.

p	q	$p \oplus q$
1	1	0
1	0	1
0	1	1
0	0	0
Chỉ đúng khi cả hai khác chân trị		

Example. “You can have tea or coffee, but not both.”

Conditional Statements

1 Propositional Logic

Definition

Let p and q be propositions. The conditional statement $p \rightarrow q$ is the proposition “if p , then q ”. The conditional statement $p \rightarrow q$ is false when p is true and q is false, and true otherwise.

In the conditional statement $p \rightarrow q$, p is called the **hypothesis** (or antecedent or premise) and q is called the **conclusion** (or consequence).

A conditional statement is also called an **implication**.

p	q	$p \rightarrow q$
1	1	1
1	0	0
0	1	1
0	0	1
Chỉ sai khi p đúng, q sai		

“if p , then q ”

“if p , q ”

“ p is sufficient for q ”

“ q if p ”

“ q when p ”

“a necessary condition for p is q ”

“ q unless $\neg p$ ”

“ p implies q ”

“ p only if q ”

“a sufficient condition for q is p ”

“ q whenever p ”

“ q is necessary for p ”

“ q follows from p ”

“ q provided that p ”

Example. “If $1+3=7$, then cats can fly.”

Biconditional statement

1 Propositional Logic

Definition

Let p and q be propositions. The biconditional statement $p \leftrightarrow q$ is the proposition “ p if and only if q ”. The biconditional statement $p \leftrightarrow q$ is true when p and q have the same truth values, and is false otherwise.

Biconditional statements are also called bi-implications.

There are some other common ways to express $p \leftrightarrow q$: “ p is necessary and sufficient for q ”, “if p then q , and conversely”, “ p iff q ”, “ p exactly when q ”.

p	q	$p \rightarrow q$	$q \rightarrow p$	$(p \rightarrow q) \wedge (q \rightarrow p) \equiv p \leftrightarrow q$
1	1	1	1	1
1	0	0	1	0
0	1	1	0	0
0	0	1	1	1
Chỉ đúng khi cả hai cùng chân trị				

Example. “ $1+1=1$ if and only if $2+2=2$ ”

Precedence of Logical Operators

1 Propositional Logic

1. Parentheses from inner to outer (dấu ”()” từ trong ra ngoài)
2. $\neg$
3. $\wedge$ and $\oplus$
4. $\vee$
5. $\rightarrow$
6. $\leftrightarrow$

Example 1. $\neg p \vee q \wedge r$ mean $(\neg p) \vee (q \wedge r)$

Example 2. Construct the truth table of the compound proposition

$$(p \vee \neg q) \rightarrow (p \wedge q).$$

Example 3. Construct the truth table of the compound proposition

$$(p \rightarrow q) \wedge (\neg p \rightarrow q) \leftrightarrow q.$$



Logic and Bit Operations

1 Propositional Logic

Definition

1. A **bit** is a symbol with two possible values, namely, 0 (zero) and 1 (one). We will use a **1** bit to represent **true** and a **0** bit to represent **false**.
2. A **Boolean variable** has value either true or false, and can be represented by a bit.
3. The operators $\neg$, $\wedge$, $\vee$ and $\oplus$ are also denoted by NOT, AND, OR and XOR.

Definition

1. A **bit string** (xâu bit) is a sequence of zero or more bits. The **length** of a bit string is the number of bits in the string.
2. The bitwise AND, OR and XOR of two strings of the same length is the string whose bits are the AND, OR and XOR of the corresponding bits of the two strings

Example 1. 101010011 is a bit string of length nine.

Example 2. Find the bitwise OR, bitwise AND, and bitwise XOR of the bit strings 01 1011 0110 and 11 0001 1101.



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Tautology and Contradiction

2 Propositional Equivalences

Definition

1. **Tautology** (hằng đúng) is a proposition that is **always true**.
2. **Contradiction** (mâu thuẫn) is a proposition that is **always false**.
3. p and q are **logical equivalent** if and only if $p \leftrightarrow q$ is a **tautology**

Note: The propositions p and q are logically equivalent if they have the same truth tables. We also write $p \equiv q$.

Example 1.

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
1	0	1	0
0	1	1	0

Example 2. Show that each of the following propositions is tautology

a. $(p \vee q) \wedge \neg p \rightarrow q$.

b. $(p \rightarrow q) \wedge (\neg p \rightarrow q) \rightarrow q$.

The Laws of Logic

2 Propositional Equivalences

Equivalence		Name
$p \wedge T \equiv p$	$p \vee F \equiv p$	Identity laws
$p \vee T \equiv T$	$p \wedge F \equiv F$	Domination laws
$p \vee p \equiv p$	$p \wedge p \equiv p$	Idempotent laws
$\neg(\neg p) \equiv p$		Double negation law
$p \vee q \equiv q \vee p$	$p \wedge q \equiv q \wedge p$	Commutative laws
$(p \vee q) \vee r \equiv p \vee (q \vee r)$	$(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$	Associative laws
$p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$	$p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$	Distributive laws
$\neg(p \vee q) \equiv \neg p \wedge \neg q$	$\neg(p \wedge q) \equiv \neg p \vee \neg q$	De Morgan's Laws
$p \vee (p \wedge q) \equiv p$	$p \wedge (p \vee q) \equiv p$	Absorption laws
$p \vee \neg p \equiv T$	$p \wedge \neg p \equiv F$	Negation laws

Example. Find the **negation** of the statement “Bob knows Python and Java.”

The Laws of Logic

2 Propositional Equivalences

Logical Equivalences Involving Conditional Statements	Logical Equivalences Involving Biconditional Statements
$p \rightarrow q \equiv \neg p \vee q$	$p \leftrightarrow q \equiv (p \rightarrow q) \wedge (q \rightarrow p)$
$p \rightarrow q \equiv \neg q \rightarrow \neg p$	$p \leftrightarrow q \equiv \neg p \leftrightarrow \neg q$
$p \vee q \equiv \neg p \rightarrow q$	$p \leftrightarrow q \equiv (p \wedge q) \vee (\neg p \wedge \neg q)$
$p \wedge q \equiv \neg(p \rightarrow \neg q)$	$\neg(p \leftrightarrow q) \equiv p \leftrightarrow \neg q$
$\neg(p \rightarrow q) \equiv p \wedge \neg q$	
$(p \rightarrow q) \wedge (p \rightarrow r) \equiv p \rightarrow (q \wedge r)$	
$(p \rightarrow r) \wedge (q \rightarrow r) \equiv (p \vee q) \rightarrow r$	
$(p \rightarrow q) \vee (p \rightarrow r) \equiv p \rightarrow (q \vee r)$	
$(p \rightarrow r) \vee (q \rightarrow r) \equiv (p \wedge q) \rightarrow r$	



Example

2 Propositional Equivalences

Show that $\neg(p \rightarrow q)$ and $p \wedge \neg q$ are logically equivalent.

Solution 1. Use a truth table.

Solution 2.

$$\begin{aligned}\neg(p \rightarrow q) &\equiv \neg(\neg p \vee q) \\ &\equiv \neg(\neg p) \wedge \neg q \\ &\equiv p \wedge \neg q\end{aligned}$$



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Predicates

3 Predicates and Quantifiers

Statements involving variables, such as

$$“x > 3”, “x = y + 3”, “x = y + z”.$$

The statement “ x is greater than 3” has two parts. The first part, the variable x , is the subject of the statement. **The second part—the predicate, “is greater than 3”.** We can denote the statement “ **x is greater than 3**” by $P(x)$, where P denotes the predicate “is greater than 3” and x is the variable. The statement $P(x)$ is also said to be the value of the propositional function P at x .

Example 1. Let $P(x)$ denote the statement “ $x > 3$.” What are the truth values of $P(4)$ and $P(2)$?

Example 2. Let $Q(x, y)$ denote the statement “ $x = y + 3$.” What are the truth values of the propositions $Q(1, 2)$ and $Q(3, 0)$?



Predicate

3 Predicates and Quantifiers

Definition

A **predicate**, or propositional function, is a function defined on a set U and returns a proposition as its value.

The set U is called the **universe of discourse**.

- We often denote a predicate by $P(x)$.
- Note that $P(x)$ is not a proposition, but $P(a)$ where a is some fixed element of U is a proposition with well determined truth value.



The universal quantifier - $\forall$

3 Predicates and Quantifiers

Definition

The **universal quantification** of $P(x)$ is the statement

“ $P(x)$ for all values of x in the domain.”

The notation $\forall x P(x)$ denotes the universal quantification of $P(x)$. Here $\forall$ is called the **universal quantifier**. We read $\forall x P(x)$ as “for all $x P(x)$ ” or “for every $x P(x)$.” An element for which $P(x)$ is false is called a counterexample to $\forall x P(x)$.

Example. Let $P(x)$ be the statement “ $x + 1 > x$ ”. What is the truth value of the quantification $\forall x P(x)$, where the domain consists of all real numbers?



The existential quantifier - $\exists$

3 Predicates and Quantifiers

Definition

The **existential quantification** of $P(x)$ is the proposition

“There exists an element x in the domain such that $P(x)$.”

We use the notation $\exists x P(x)$ for the existential quantification of $P(x)$. Here $\exists$ is called the **existential quantifier**.

Example. Let $P(x)$ denote the statement “ $x > 3$.” What is the truth value of the quantification $\exists x P(x)$, where the domain consists of all real numbers?

Quantifiers

3 Predicates and Quantifiers

Statement	When True?	When False?
$\forall x P(x)$	$P(x)$ is true for every x .	There is an x for which $P(x)$ is false.
$\exists x P(x)$	There is an x for which $P(x)$ is true.	$P(x)$ is false for every x .

Precedence of Quantifiers.

The quantifiers $\forall$ and $\exists$ have **higher precedence than all logical operators** from propositional calculus.

For example, $\forall x P(x) \vee Q(x)$ is the disjunction of $\forall x P(x)$ and $Q(x)$.



Example

3 Predicates and Quantifiers

Suppose the universe of discourse is all creatures, and define the following:

- $L(x)$ = “x is a lion.”
- $F(x)$ = “x is fierce.”
- $C(x)$ = “x drinks coffee.”

1. All lions are fierce $\rightarrow \forall x(L(x) \rightarrow F(x))$
2. Some lions don't drink coffee $\rightarrow \exists x(L(x) \wedge \neg C(x))$
3. Some fierce creatures don't drink coffee $\rightarrow \exists x(F(x) \wedge \neg C(x))$

The Order of Quantifiers

3 Predicates and Quantifiers

Example. $P(x,y)$ = “x is sitting next to y”

TABLE 1 Quantifications of Two Variables.		
Statement	When True?	When False?
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

- $\forall x \forall y P(x, y)$ - everyone is sitting next to everyone else. FALSE
- $\exists x \exists y P(x, y)$ - there are two people sitting next to each other TRUE
- $\forall x \exists y P(x, y)$ - every person is sitting next to somebody TRUE
- $\exists x \forall y P(x, y)$ - a particular person is sitting next to everyone else FALSE

Negating Quantified Expressions

3 Predicates and Quantifiers

Rule.

$$\overline{\forall x P(x)} \equiv \exists x \overline{P(x)}.$$

and

$$\overline{\exists x P(x)} \equiv \forall x \overline{P(x)}.$$

Example 1. What are the negations of the statements $\forall x(x^2 > x)$ and $\exists x(x^2 = 2)$?

Example 2.

$$\begin{aligned} \neg \forall x \exists y (xy = 1) &\equiv \exists x \neg \exists y (xy = 1) \\ &\equiv (\exists x)(\forall y) \neg (xy = 1) \\ &\equiv (\exists x)(\forall y)(xy \neq 1) \end{aligned}$$



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Introduction

4 Rules of Inference

Consider the following argument involving propositions (which, by definition, is a sequence of propositions):

“If you have a current password, then you can log onto the network.”

“You have a current password.”

Therefore,

“You can log onto the network.”

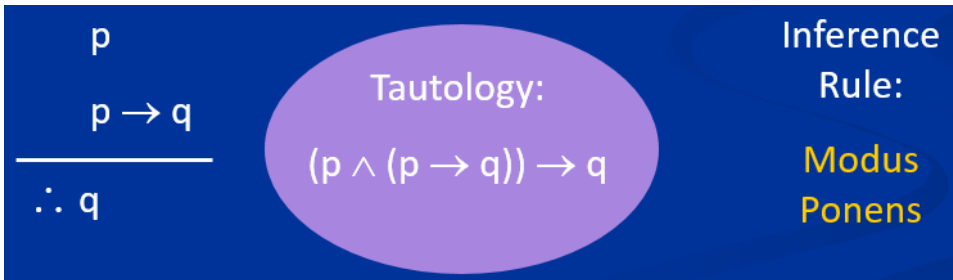
Use p to represent “You have a current password” and q to represent “You can log onto the network.” Then, the argument has the form

$$\frac{p \rightarrow q}{p} \therefore q$$

where $\therefore$ is the symbol that denotes “therefore.”

Modus Ponens (khẳng định)

4 Rules of Inference



Example.

If I am Duong, then I am a Mathematics Lecturer.

I am Duong.

$\therefore$ I am a Mathematics Lecture.

Modus Tollens (phủ định)

4 Rules of Inference

$\frac{\neg q \quad p \rightarrow q}{\therefore \neg p}$	Tautology: $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$	Inference Rule: Modus Tollens
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Example.

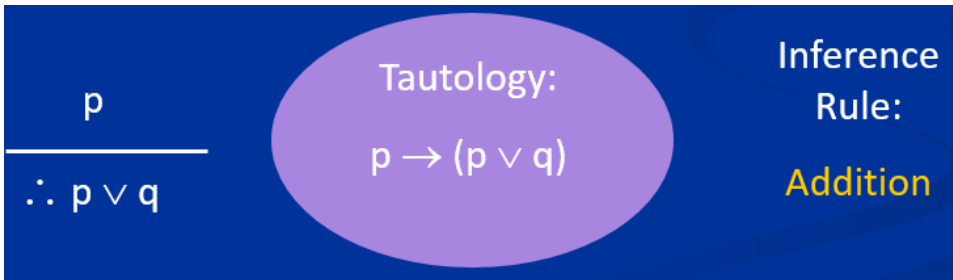
The sum of the interior angles of this figure is not 180° .

If this figure is a triangle, then the sum of its interior angles is 180° .

$\therefore$ This figure is not a triangle.

Addition

4 Rules of Inference



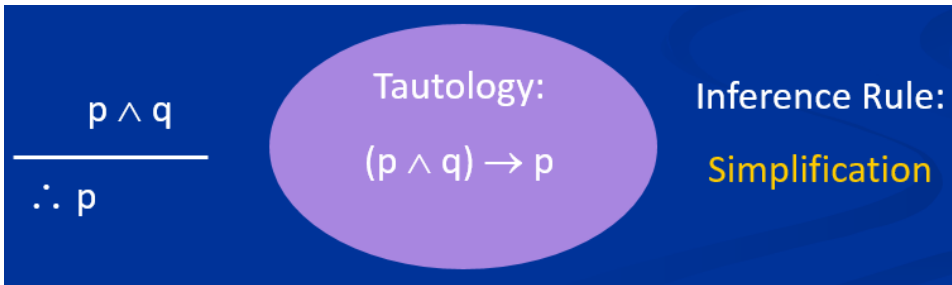
Example.

I am not a great football striker.

$\therefore$ I am not a great football striker or I am tall.

Simplification

4 Rules of Inference



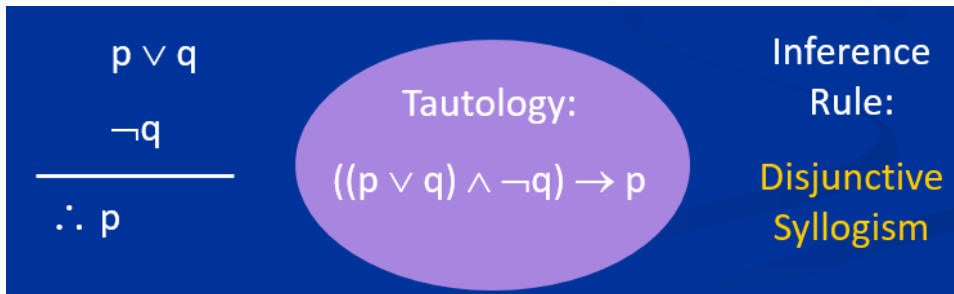
Example.

Sandra knows Java and Sandra knows C++.

$\therefore$ Sandra knows C++.

Disjunctive Syllogism (loại trừ)

4 Rules of Inference



Example.

I am teacher or doctor.

I am not teacher.

$\therefore$ I am doctor.

Hypothetical Syllogism (tam đoạn luận)

4 Rules of Inference

$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$	Tautology: $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$	Inference Rule: Hypothetical Syllogism
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Example.

If it rains, then the ground gets wet.

If the ground gets wet, then the streets become slippery.

$\therefore$ If it rains, then the streets become slippery.

Fallacies

4 Rules of Inference

$$\begin{aligned}14 + 6 - 20 &= 21 + 9 - 30 \\ \implies 2(7 + 3 - 10) &= 3(7 + 3 - 10) \\ \implies 2 &= 3\end{aligned}$$

- Rules of inference, appropriately applied give **valid** arguments.
- Mistakes in applying rules of inference are called **fallacies** (ngụy biện).

Example 1.

If I am Descartes, then I am a mathematician.

I am a mathematician.

$\therefore$ I am Descartes.

Example 2.

If it rains then it is cloudy.

It does not rain.

$\therefore$ It is not cloudy.

Review

4 Rules of Inference

Some rules of inference

Modus Ponens

$$\begin{array}{l} p \\ p \rightarrow q \\ \hline \therefore q \end{array}$$

Tautology:
 $(p \wedge (p \rightarrow q)) \rightarrow q$

Modus Tollens

$$\begin{array}{l} \neg q \\ p \rightarrow q \\ \hline \therefore \neg p \end{array}$$

Tautology:
 $(\neg q \wedge (p \rightarrow q)) \rightarrow \neg p$

Addition

$$\begin{array}{l} p \\ \hline \therefore p \vee q \end{array}$$

Tautology:
 $p \rightarrow (p \vee q)$

Simplification

$$\begin{array}{l} p \wedge q \\ \hline \therefore p \end{array}$$

Tautology:
 $(p \wedge q) \rightarrow p$

Disjunctive Syllogism

$$\begin{array}{l} p \vee q \\ \neg q \\ \hline \therefore p \end{array}$$

Tautology:
 $((p \vee q) \wedge \neg q) \rightarrow p$

Hypothetical Syllogism

$$\begin{array}{l} p \rightarrow q \\ q \rightarrow r \\ \hline \therefore p \rightarrow r \end{array}$$

Tautology:
 $((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$

Quantified Statements

4 Rules of Inference

Universal Instantiation. If $\forall xP(x)$ is true, then $P(c)$ is true for any choice of c in the universe of discourse.

$$\frac{\forall xP(x) \text{ is true.}}{\therefore P(c) \text{ is true.}}$$

Universal generalization. If $P(c)$ is true for any choice of c in the universe of discourse, then $\forall xP(x)$ is true.

$$\frac{P(c) \text{ is true for any choice of } c.}{\therefore \forall xP(x) \text{ is true.}}$$

Example.

All humans are mortal.

Socrates is human.

Therefore, Socrates is mortal.

$p(x)$: "x is human.", $q(x)$: "x is mortal."

$\forall x, p(x) \rightarrow q(x)$.

$p(\text{Socrates})$

Therefore, $q(\text{Socrates})$



Quiz

4 Rules of Inference

Q1.

Premise: Everyone, who is over 30, can read newspaper.

Premise: Mr. Bean can read newspaper.

Premise: Mrs. Bean is 36 years old.

Conclusion 1: He is over 30.

Conclusion 2: She can read newspaper.

Select one:

- a. The conclusion 1 is logical and the conclusion 2 is not.
- b. Both conclusions are logical.
- c. The conclusion 2 is logical and the conclusion 1 is not.
- d. Both conclusions are illogical.

Q2.

Study the following arguments:

(i) If Mr. Bean can speak English, then he is good. he can't speak English. Therefore, he is not good.

(ii) All Cr7 fans love FC MU. Mr. Bean doesn't love FC MU. Therefore, he is not a CR7 fan.

(i) is ... and (ii) is ...

Select one:

- a. illogical, illogical
- b. illogical, logical
- c. logical, illogical
- d. logical, logical



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Problems

5 Problems

1. Let p and q be the propositions.

p : It is below freezing.

q : It is snowing.

Write these propositions using p and q and logical connectives.

a) It is below freezing and snowing.

b) It is below freezing but not snowing.

c) It is not below freezing and it is not snowing.

d) It is either snowing or below freezing (or both).

2. Find the bitwise OR, bitwise AND, and bitwise XOR.

a) 101 1110, 010 0001

b) 1111 0000, 1010 1010

3. Determine whether each of the following arguments is valid or not valid.

a) All parrots like fruit. My pet bird is not a parrot. Therefore, my pet bird does not like fruit.

b) Everyone who eats granola every day is healthy. Linda is not healthy. Therefore, Linda does not eat granola every day.



Q&A

Thank you for listening!